2022 Ph H2 Q9

Section: Particles and Waves

Topic: Photoelectric Effect

Photoemission from metal X studied. (a) Determine threshold frequency from graph. (b) Identify metal X from table of work functions, with justification.

Worked solution

(a) Threshold frequency

From the graph, extrapolation to Ek = 0 gives $f_0 \approx 5.0\times10^{14}$ Hz.

Answer: $f_0 = 5.0\times10^{14}$ Hz

(b) Identifying metal

Work function $\varphi = h f_0$
$= (6.63\times10^{-34})(5.0\times10^{14})$
= 3.31e-19 J

Compare with data:
Potassium: 3.7×10^{-19} J
Calcium: 4.6×10^{-19} J
Zinc: 5.8×10^{-19} J
Gold: 8.5×10^{-19} J

Calculated φ = 3.31e-19 J, closest to Calcium (4.6×10^{-19} J).

Answer: Metal X is Calcium

Final answers

(a) $f_0 = 5.0\times10^{14}$ Hz

(b) Metal X = Calcium ($\varphi \approx 4.6\times10^{-19}$ J, matches calculation)

Revision tips

- Threshold frequency: minimum f needed for photoemission.
- Work function $\varphi = h f_0$.
- Graph of Ek vs f: straight line slope = h, intercept = $-\varphi$.
- Match calculated φ with data table to identify the metal.